

Fig. 2: Actual-size PCB of the motion detector

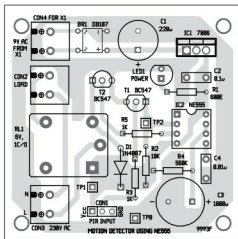


Fig. 3: Component layout of the PCB

man body infrared radiations. It has a single output that goes high when a valid motion is detected. That is, the load is switched on whenever the PIR module senses a body in motion nearby.

Under normal conditions, transistor T1 is cut off and collector output is high around 6V. When motion is

sensed, output pin (OUT) of the sensor becomes high, making transistor T1 to conduct for a few seconds, and voltage at its collector goes low momentarily. This low signal triggers IC2. Output pin 3 of IC2 goes high for around ten minutes, energising relay RL1 through transistor T2, turning on the load for ten minutes.

In brief, when someone comes in front of the PIR module, its output triggers IC2 to turn on the load. Thereafter it is disabled automatically.

Construction and testing

An actual-size, single-side PCB for the motion detector using NE555 timer is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a box and install it at a suitable location. Connect the PIR module across CON1.

Connect transformer X1 to 230V AC mains. Verify various test point voltages as given in the table to ensure proper working before using the circuit. ●



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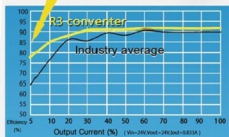
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